MODULE 2 UPPER EXTREMITY

The Upper Extremity.

APPENDICULAR SKELETON

The pectoral girdle and the bones of the arm, forearm, and hand. This chapter also carries the bone markings vocabulary, which repeats on every bone for the rest of the course.

BY THE END

1. Name the two bones of the pectoral girdle and explain how the girdle attaches the upper limb to the axial skeleton.
2. Trace the bones of the upper limb from shoulder to fingertips.
3. Tell the radius from the ulna by side and by the markings at each end.
4. Name the three groups of hand bones and how many bones are in each.
5. Use the bone markings vocabulary to classify any landmark as a projection, a depression, or an opening.

WATCH AFTER YOU READ THIS CHAPTER

Concept videos: Appendicular Skeleton, Upper

drsrennie-stack.github.io/new-build-bio4-solano/appendicular-concept-videos.html

Bone Markings Vocabulary

Learn this before the landmarks. Every marking is a projection, a depression, or an opening, and these terms repeat on every bone you will study.

TYPE, DESCRIPTION, EXAMPLE

Marking	Type	Description	Example
Process	Projection	Any bony prominence	Mastoid process
Tubercle	Projection	Small rounded bump	Greater tubercle of the humerus
Tuberosity	Projection	Large rough projection	Deltoid tuberosity
Head	Projection	Bony expansion carried on a neck	Head of the humerus
Epicondyle	Projection	Raised area above a condyle	Medial epicondyle of the humerus
Facet	Articular surface	Smooth, nearly flat surface	Articular facets of vertebrae
Fossa	Depression	Shallow basin	Olecranon fossa
Sulcus	Depression	A groove	Intertubercular sulcus
Foramen	Opening	Hole for vessels or nerves	Nutrient foramen

The Upper Extremity, an Overview

The appendicular skeleton is the bones of the limbs and the girdles that attach them to the axial skeleton.

FOUR PARTS, PROXIMAL TO DISTAL

Part	Region it spans	Bones
Pectoral girdle	Attaches the upper limb to the trunk	Clavicle and scapula
Arm	Shoulder to elbow	Humerus
Forearm	Elbow to wrist	Radius and ulna
Hand	Wrist to fingertips	Carpals, metacarpals, and phalanges

The Pectoral Girdle

Clavicle

MARKINGS

Feature	What it is
Clavicle	The collarbone, an S-shaped bone across the top of the thorax
Sternal end	The medial end. Articulates with the manubrium of the sternum
Acromial end	The lateral end. Articulates with the acromion of the scapula
The only bony link	The clavicle is the sole bony attachment of the upper limb to the axial skeleton

Scapula

MARKINGS

Feature	What it is
Scapula	The shoulder blade, a flat triangular bone on the posterior thorax
Borders	The superior, medial, and lateral borders of the triangle
Spine	The ridge running across the posterior surface
Acromion	The flat process at the tip of the spine, the point of the shoulder
Coracoid process	The hook-like anterior process, a muscle attachment
Glenoid cavity	The shallow socket that receives the head of the humerus
Supraspinous fossa	The hollow above the spine
Infraspinous fossa	The hollow below the spine
Subscapular fossa	The hollow on the anterior surface

Joints of the pectoral girdle

THREE JOINTS

Joint	What meets what
Sternoclavicular joint	Sternal end of the clavicle with the manubrium. The only joint linking the upper limb to the axial skeleton
Acromioclavicular joint	Acromial end of the clavicle with the acromion of the scapula
Glenohumeral joint	Head of the humerus in the glenoid cavity. The shoulder joint

The Arm and Forearm

Humerus

MARKINGS, PROXIMAL TO DISTAL

Marking	What it is
Humerus	The single bone of the arm, from shoulder to elbow
Head	The rounded proximal end that fits into the glenoid cavity
Anatomical neck	The narrow groove just below the head
Surgical neck	The narrowing below the tubercles. A common fracture site
Greater tubercle	The lateral bump near the head, a muscle attachment
Lesser tubercle	The anterior bump near the head, a muscle attachment
Intertubercular sulcus	The groove between the tubercles that holds a biceps tendon
Deltoid tuberosity	The roughened ridge on the shaft where the deltoid muscle attaches
Capitulum	The lateral knob at the distal end. Articulates with the radius
Trochlea	The medial spool at the distal end. Articulates with the ulna
Medial epicondyle	The medial projection above the trochlea
Lateral epicondyle	The lateral projection above the capitulum
Olecranon fossa	The deep posterior pit that receives the ulna when the elbow straightens

HOLD ONTO THIS

Capitulum sits lateral and meets the radius. Trochlea sits medial and meets the ulna. Both pairings hold in anatomical position, so fix the position first and the pairing follows.

The forearm bones, radius and ulna

SIDE AND MARKINGS AT EACH END

Bone	Side of the forearm	Proximal (elbow) end	Distal (wrist) end
Radius	Lateral, the thumb side	Disc-shaped head articulates with the capitulum. Neck just below. Radial tuberosity, a muscle attachment, below the neck	Styloid process on the lateral side. Ulnar notch on the medial side receives the head of the ulna, forming the distal radioulnar joint
Ulna	Medial, the little-finger side	Olecranon, the point of the elbow. Coronoid process in front. Trochlear notch grips the trochlea of the humerus. Radial notch on the lateral side receives the head of the radius, forming the proximal radioulnar joint	Styloid process on the medial side

Grooves, nerves, and the joining membrane

WHAT RUNS BETWEEN AND ALONG THE BONES

Structure	What it is
Radial groove	An oblique groove on the posterior shaft of the humerus. The radial nerve runs in it
Ulnar nerve at the medial epicondyle	The nerve passes just behind the medial epicondyle, where it can be palpated. The funny bone
Interosseous membrane	A fibrous sheet, a syndesmosis, spanning the shafts of the radius and ulna. A muscle attachment that also binds the two bones
Three radioulnar joints	Proximal (radial head and radial notch), distal (ulnar head and ulnar notch), and the interosseous membrane between the shafts

The Hand

Carpals, the wrist

EIGHT BONES IN TWO ROWS

Group	Bones
Carpals	Eight small bones arranged in two rows, forming the wrist
Proximal row	Scaphoid, lunate, triquetrum, and pisiform
Distal row	Trapezium, trapezoid, capitate, and hamate

MNEMONIC

So Long To Pinky, Here Comes The Thumb.

Metacarpals and phalanges

THE PALM AND THE DIGITS

Structure	What it is
Metacarpals	Five bones forming the palm, numbered 1 to 5 starting at the thumb
Phalanges	The finger bones, fourteen in each hand
Thumb	The pollex. Has two phalanges, a proximal and a distal
Fingers	Each of the other four has three phalanges: proximal, middle, and distal

The carpal tunnel

ROOF, CONTENTS, AND WHAT GOES WRONG

Structure	What it is
Flexor retinaculum	A strong fibrous band roofing the carpal bones. The space beneath it is the carpal tunnel
Contents	The long flexor tendons of the fingers and thumb, and the median nerve
Carpal tunnel syndrome	Narrowing of the tunnel compresses the median nerve
Clinically notable carpals	Capitate is the largest. Scaphoid is the most frequently fractured. Lunate is the most frequently dislocated

THE BONES THAT BREAK WHEN YOU FALL ON AN OUTSTRETCHED HAND

The clavicle is the most commonly broken bone in the body, usually in its middle third. It is the slender strut that takes the force when you fall onto a shoulder or an outstretched arm. The surgical neck of the humerus fractures often, and the axillary nerve runs close by, so a break there is checked for nerve injury. The scaphoid is the carpal bone most often fractured, again from a fall on an outstretched hand, and its blood supply is poor, so a missed scaphoid fracture may fail to heal. Tenderness in the anatomical snuffbox, the hollow at the base of the thumb, is the clue. The ulnar nerve passes behind the medial epicondyle of the humerus, the spot people call the funny bone, which is why a knock there sends a jolt to the little finger.